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**B.TECH. IV SEM MAIN EXAM
AUGUST-2023**

**COMPUTER SCIENCE AND ENGINEERING
(4CS4-03) - THEORY OF COMPUTATION**

Common to CS,IT,AI, DS,MC,CM,CD,CA,AD,AM,CY,IO

Time : 3 Hours]

[Max. Marks : 70

[Min Passing Marks : 20]

Instructions to Candidates : Part – A : Short answer type questions (up to 25 words)

10 × 2 marks = 20 marks. All ten questions are compulsory.

Part – B: Analytical/Problem Solving questions 5 × 4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

PART A

Q. 1. Formalling Define Deterministic finite automata.

- Q. 2. CFG is ambiguous. Justify.
- Q. 3. Define the language recognised by Turing Machine.
- Q. 4. Compare transitions functions of NFA and DFA.
- Q. 5. Can we use finite state automata to evaluate 1's complement of a binary numbers ?
- Q. 6. Write notations for language accepted by DFA and NFA.
- Q. 7. List some of applications of automata Theory.
- Q. 8. Give a regular expression for the set of all strings not containing 101 as a substring.
- Q. 9. What do you Mean by closure properties of regular languages ? Define some important closer properties.
- Q. 10. Prove that Regularset are closed over intersection Operation.

PART B

- Q. 1. Design a PDA that accepts the language $L = [a^n b^m a^n : m, n > 0]$.
- Q. 2. Describe vertex cover problem with example.
- Q. 3. Convert the following grammer G into GNF :
- $$S \rightarrow XA \mid BB$$
- $$B \rightarrow b \mid SB$$
- $$X \rightarrow b$$
- $$A \rightarrow a$$
- Q. 4. Construct a turing machine for checking the palindrome of a string of odd palindrome for $\Sigma = \{0, 1\}$.
- Q. 5. Is it true that deterministic PDA and non-deterministic PDA are equivalent in the sense of language of acceptances ? Justify
- Q. 6. Differentiate between tractable and untractable problems.
- Q. 7. Explain the pumping Lemma of CFL.

PART C

- Q. 1. Explain/highlight the implications of halting problems.
- Q. 2. Define derivation tree. Explain its uses with example.
- Q. 3. Write Chomsky hierarchy of languages. Prepare a table indicating the automata and grammars for languages in chomsky Hierarchy.
- Q. 4. Write a short note on Universal Turing Machine.
- Q. 5. Discuss about Multitape Turing Machines. Explain informally how they can simulate the moves of a turing machine.
